Testing phylogenetic signal under rate heterogeneity and invariant-site mixtures with SatuTe

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Abstract

Sequence saturation can leave a branch in a phylogeny without enough signal to justify the split it represents. SatuTe tests this by estimating a coherence coefficient for the second largest eigenvalue of the substitution model. We ask whether, under rate-heterogeneous Tree-of-Life models, the remaining non-stationary directions should contribute to the same test, and how an explicit invariant-site component changes the infinite-branch null. The relative eigenvalue-weighted statistic keeps the SatuTe null hypothesis and reduces to SatuTe under the Jukes–Cantor model. For $+I$ mixtures, the invariant category remains jointly distributed across an infinite focal branch and must enter only the null denominator. In a dense paired $+I+G4/+I+R4$ grid of 10.8 million simulated alignments, the refined invariant-null statistic gave a mean nominal false-positive rate of 4.92%, compared with 90.0% for the counterfactual construction. After independent null calibration, the largest power gains were 11.6 percentage points for $+I+G4$ and 16.3 percentage points for $+I+R4$, localized to the detection boundary. The refinement therefore corrects a structural $+I$ null mismatch and can modestly improve boundary power; it does not change $+G$-only analyses.

Saturation occurs when repeated substitutions erase phylogenetic information from an alignment. For two sequences, saturation means that the alignment no longer justifies their inferred evolutionary separation. In larger alignments, the same problem appears at the level of branches. A branch can appear in a maximum-likelihood tree, and can have high conventional support, while the alignment carries little signal for the two groups separated by that branch.

SatuTe generalises saturation between two sequences to a test of saturation between subtrees [1]. A branch AB splits a site pattern ∂ into subpatterns ∂_A and ∂_B on the subtrees T_A and T_B. The null hypothesis is that the two subtrees evolve independently. Rejection implies phylogenetic signal for branch AB; otherwise, the branch is considered saturated.

The test uses the spectral decomposition of the reversible substitution model [2]. For a site pattern ∂, SatuTe computes partial likelihood vectors $L(\partial_A)$ and $L(\partial_B)$. It also computes subtree pattern probabilities $P(\partial_A) = \langle L(\partial_A), \pi\rangle$ and $P(\partial_B) = \langle L(\partial_B), \pi\rangle$. With eigenvalues $\lambda_0 = 0 > \lambda_1 \geq \lambda_2 \geq \cdots$ and left eigenvectors h_i, the coherence coefficient for a simple eigenvalue is

$$C_{\partial i} = \left\langle \frac{L(\partial_A)}{P(\partial_A)}, h_i \right\rangle \left\langle \frac{L(\partial_B)}{P(\partial_B)}, h_i \right\rangle. \tag{1}$$

SatuTe uses this coefficient for the second largest eigenvalue. Across n sites, the empirical statistic is $\hat{C}_1 = n^{-1}\sum_{s=1}^{n} C_{\partial_s 1}$, with variance estimator $\hat{\sigma}_1^2$. The one-sided statistic is $Z = \sqrt{n}\hat{C}_1/\hat{\sigma}_1$. The test rejects subtree independence when $Z > z_\alpha$.

The second largest eigenvalue gives the slowest non-stationary decay towards the stationary distribution. It is therefore the conservative component on long branches, because all other non-stationary components decay faster. Under general time-reversible nucleotide models [3] and amino-acid models such as LG, these non-stationary eigenvalues are usually distinct. We ask whether including the faster components changes which branches or alignment regions are called phylogenetically informative when a fitted rate-heterogeneous model separates them.

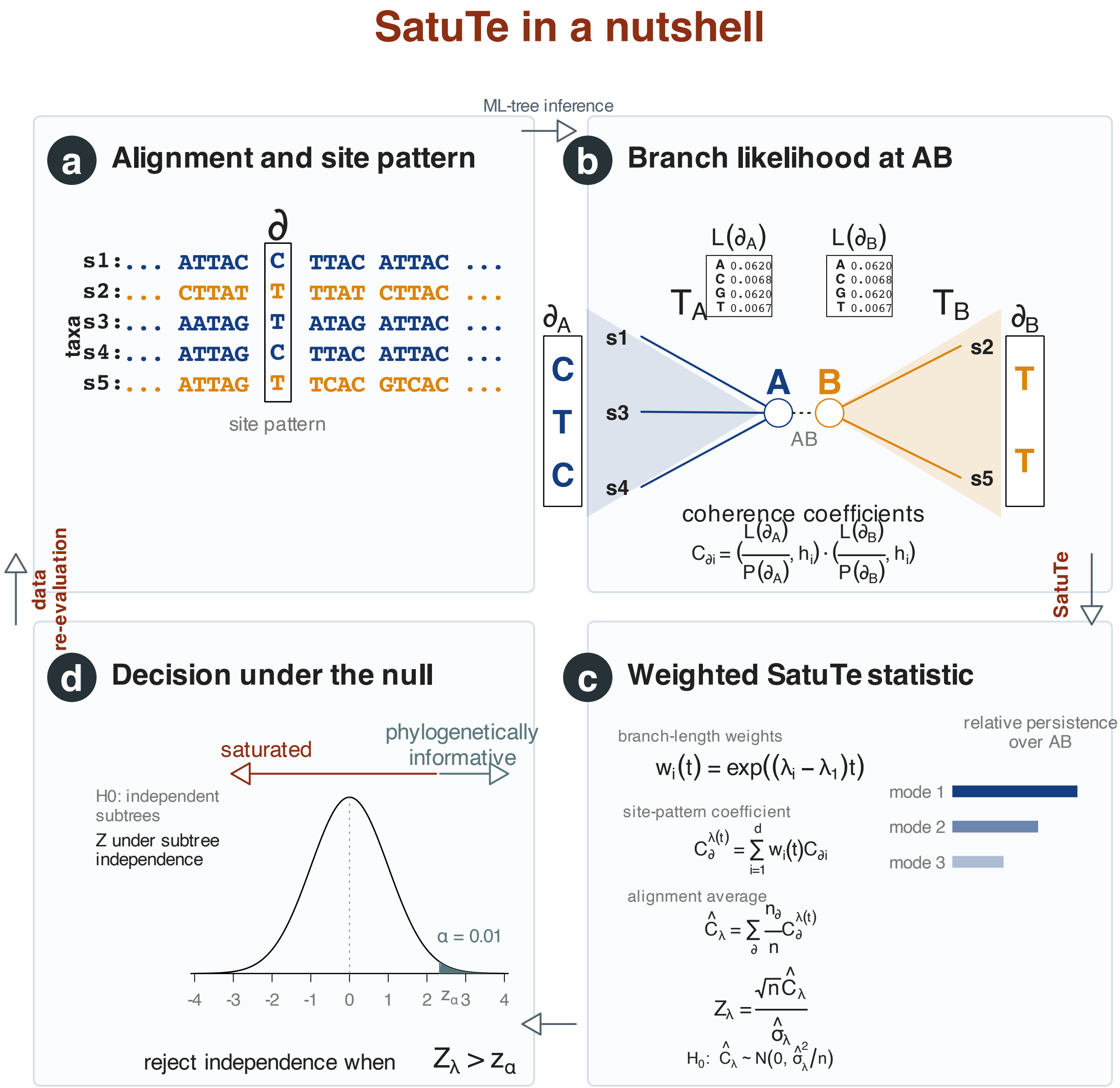

Figure 1: **SatuTe in a nutshell. a,** Five-taxon DNA alignment where each column represents a pattern ∂. **b,** The branch AB splits the five-taxon tree into subtree T_A with sequences $S1, S3, S4$ and subtree T_B with sequences $S2, S5$. Branch AB also splits each pattern, such as $\partial = \text{CTTCT}$, into subpatterns $\partial_A = \text{CTC}$ and $\partial_B = \text{TT}$. These subpatterns are used to compute the likelihood vectors $L(\partial_A)$ and $L(\partial_B)$, which determine the coherence coefficients. **c,** The weighted statistic used here keeps the SatuTe coherence coefficients and weights each non-stationary component by its relative persistence over branch length t. **d,** The null hypothesis of independently evolving subtrees is rejected at the significance level α if $Z_\lambda > z_\alpha$. Rejection implies phylogenetic signal for branch AB.

Results

A weighted coefficient keeps the SatuTe test

The spectral decomposition used by SatuTe gives

$$\frac{P(\partial \mid t)}{P(\partial_A)P(\partial_B)} = 1 + \sum_{i=1}^{d} C_{\partial i} \exp(\lambda_i t). \tag{2}$$

Here d is the number of non-stationary eigenvalues: $d = 3$ for DNA and $d = 19$ for an amino-acid model. The coefficient $C_{\partial i}$ measures agreement between the two subtrees in spectral direction h_i. The factor $\exp(\lambda_i t)$ measures how much of that direction remains after evolution across branch AB. Because all non-stationary eigenvalues are negative, components with more negative λ_i decay faster.

We define the eigenvalue-weighted coherence for pattern ∂ as

$$C_\partial^\lambda(t) = \sum_{i=1}^{d} w_i(t) C_{\partial i}, \qquad w_i(t) = \exp\{(\lambda_i - \lambda_1)t\}. \tag{3}$$

The relative weight $w_i(t)$ sets $w_1(t) = 1$ and downweights components that decay faster than the second-largest-eigenvalue component. Thus the published SatuTe coefficient remains the reference scale, while the other components contribute according to the fitted branch length.

For an alignment of n independently evolving sites, the empirical coefficient is

$$\hat{C}_\lambda(t) = \frac{1}{n} \sum_{s=1}^{n} C_{\partial_s}^\lambda(t). \tag{4}$$

The variance estimator has the same form as the SatuTe estimator for a repeated second largest eigenvalue, with the mode weights included:

$$\hat{\sigma}_\lambda^2(t) = \sum_{i=1}^{d} \sum_{j=1}^{d} w_i(t) w_j(t) \hat{\sigma}_{A,i,j}^2 \hat{\sigma}_{B,i,j}^2. \tag{5}$$

The corresponding test statistic is

$$Z_\lambda(t) = \frac{\sqrt{n}\hat{C}_\lambda(t)}{\hat{\sigma}_\lambda(t)}. \tag{6}$$

The decision rule is unchanged. Branch AB is called phylogenetically informative when $Z_\lambda(t) > z_\alpha$. When the tree is inferred from the same alignment, the same Bonferroni correction used by SatuTe gives the threshold $z_{\alpha/(m_A m_B)}$.

Invariant categories require a different null likelihood

An explicit invariant category is a boundary case of the rate mixture. Let q_c and r_c be the proportion and rate of category c, and let A_{sc} and B_{sc} be the two subtree-pattern probabilities for site pattern s. For a positive-rate category, the infinite-branch null contribution is $q_c A_{sc} B_{sc}$. For the invariant category $c = 0$, however, $r_0 = 0$ and the transition matrix remains the identity even as the focal branch length tends to infinity. Its null contribution is therefore the joint probability

$$J_{s0} = \sum_a \pi_a L_{A,s0}(a) L_{B,s0}(a), \tag{7}$$

not the product $A_{s0}B_{s0}$. The correct mixture null is

$$p_\infty(s) = q_0 J_{s0} + \sum_{c:r_c>0} q_c A_{sc} B_{sc}. \tag{8}$$

For the positive-rate categories, define the model-weighted site score

$$u_s(t) = \frac{\sum_{c:r_c>0} q_c A_{sc} B_{sc} \sum_{i=1}^{d} e^{\lambda_i r_c t} C_{sc,i}}{p_\infty(s)}. \tag{9}$$

The invariant component enters only the denominator and contributes no finite-versus-infinite-branch signal. This gives the exact identity

$$\frac{p_t(s)}{p_\infty(s)} = 1 + u_s(t). \tag{10}$$

In computation, all positive-rate persistence weights are divided by one common factor e^{δ}, where $\delta = \max_{c:r_c>0,i} \lambda_i r_c t$. This prevents underflow without changing the studentized statistic. The refined test uses the sample mean and sample variance of these globally rescaled site scores. We refer to it as the refined invariant-null statistic.

The previous counterfactual construction treated the invariant category as if it became independent across an infinite focal branch, replacing J_{s0} by $A_{s0}B_{s0}$ and retaining a zero-rate numerator contribution. That construction does not satisfy Eq. (10) under a $+I$ model and need not have null expectation zero.

The Jukes–Cantor case is unchanged

The Jukes–Cantor (JC) model has one stationary eigenvalue and a three-dimensional non-stationary eigenspace with $\lambda_1 = \lambda_2 = \lambda_3$. Therefore $w_i(t) = 1$ for all non-stationary modes in Eq. (3). The weighted statistic becomes

$$C_\partial^\lambda(t) = \sum_{k=1}^{3} C_{\partial 1,k} = C_{\partial 1}, \tag{11}$$

which is the multiplicity-three expression used by SatuTe. Thus the weighted statistic does not change the JC statistic. If the substitution model does not distinguish among non-stationary directions, the weighted statistic does not introduce that distinction.

The original SatuTe simulations define the baseline

The simulations follow the two topologies used in the original SatuTe paper. In the five-taxon case, the tested external branch AB connects taxon B to the root A of a balanced four-taxon subtree T_A. All branches inside T_A have length 0.2, and the length of AB is varied. In the 16-taxon case, AB connects the roots A and B of two balanced eight-taxon subtrees. Branch lengths inside these subtrees are sampled from EvoNAPS-derived empirical branch-length distributions in the interval 0.04 to 0.2. Each simulated alignment is analyzed under four settings. Two settings use the true topology, with either fixed true branch lengths or maximum-likelihood branch lengths. The other two use a maximum-likelihood inferred tree, without or with the Bonferroni correction used by SatuTe.

Rate-heterogeneous simulations use full alignments

The rate-heterogeneous comparison uses full alignments, not sliding windows. Each replicate is evaluated twice on the same tree, branch length, target split and random seed. One evaluation uses the published SatuTe coefficient $C_{\partial 1}$, and the other uses the eigenvalue-weighted coefficient $C_\partial^\lambda(t)$. The paired design isolates the effect of including the additional non-stationary components.

The first grid used 100 replicates per condition and increased the alignment length from 100 to 10,000 sites. The second grid used 1,000 replicates per condition for 100, 250 and

1,000 sites on the same five-taxon external and 16-taxon internal target branches. Both grids used Tree-of-Life-derived four-category rate-heterogeneous models. The nucleotide setting was GTR+F+G4 fitted to the 16S rRNA Tree-of-Life example, and the protein setting was LG+G4 fitted to the Tree-of-Life protein example.

The corrected invariant null restores false-positive control

The dense invariant-mixture grid evaluated 270 model–branch–alignment-length cells. With 20,000 null and 20,000 alternative replicates per cell, 10.8 million paired alignments were evaluated by both formulas. Exact enumeration verified Eq. (10) with maximum absolute error 1.55×10^{-11}. The refined score had exact null expectation below 1.53×10^{-16} in absolute value. The counterfactual score retained a positive null expectation between 0.0354 and 0.2176 across the branch grid.

At the ordinary one-sided $N(0,1)$ threshold, the counterfactual null rejection rate averaged 0.900 across the 270 cells and ranged from 0.500 to 1.000. The refined rejection rate averaged 0.0492 and ranged from 0.0423 to 0.0591. Splitting the null sample into threshold-training and held-out halves restored empirical rejection rates near 0.05 for both formulas, with means of 0.0501 and 0.0499, respectively. This calibration did not repair the counterfactual statistic as a standard-normal pivot: at $t = 10{,}000$, its empirical threshold increased from approximately 3.15 at 100 sites to 18.47 at 10,000 sites. The refined threshold remained between 1.62 and 1.69.

The refined invariant statistic gains power at the detection boundary

After independent null calibration, the refined statistic had the larger exact null-centered standardized separation in 245 of 270 cells and equal separation in the remaining 25 extreme-saturation cells; it was never lower. The observed paired power difference was concentrated in a diagonal ridge: longer alignments moved the detection boundary to longer branches. The largest gain for each alignment length was:

Model	Sites	Branch t	Counterfactual	Refined	Difference
$+I + G4$	100	50	0.100	0.115	0.015
	500	40	0.342	0.384	0.042
	1,000	45	0.406	0.456	0.050
	5,000	60	0.482	0.584	0.102
	10,000	60	0.718	0.835	0.116
$+I + R4$	100	10	0.558	0.573	0.015
	500	20	0.405	0.450	0.044
	1,000	22.5	0.449	0.522	0.073
	5,000	30	0.501	0.605	0.104
	10,000	35	0.402	0.565	0.163

Table 1: **Largest calibrated power gain at each alignment length in the dense invariant-mixture grid.** Differences are refined minus counterfactual. Every alignment is evaluated by both formulas.

At the largest $+I + R4$ gain, $t = 35$ and 10,000 sites, the paired power difference was 0.163 with bootstrap 95% interval $[0.157, 0.170]$. The largest $+I + G4$ gain was 0.116 at $t = 60$ and 10,000 sites, with interval $[0.111, 0.122]$. The most negative observed difference among all 270 cells was -0.0059. These sub-percentage-point losses occurred mainly where both calibrated powers were near the null floor; with 270 unadjusted cell-wise comparisons they are compatible with Monte Carlo threshold variation and were not supported by a lower exact standardized separation.

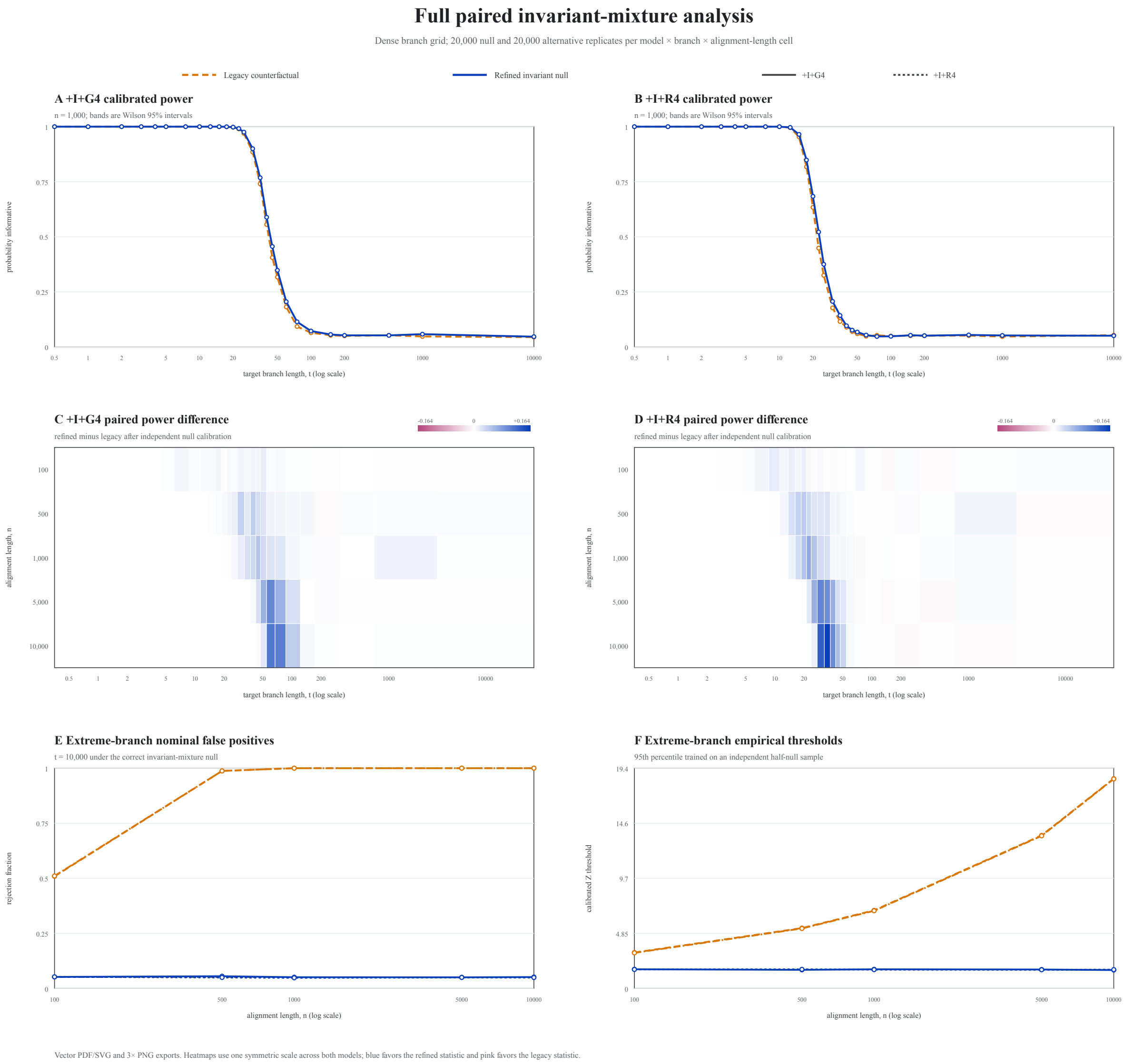

Figure 2: **Full paired $+I+G4$ and $+I+R4$ invariant-mixture analysis. a,b,** Formula-specific empirically calibrated power at 1,000 sites across the 27-step branch grid; bands are Wilson 95% intervals. **c,d,** Refined-minus-counterfactual calibrated power for all five alignment lengths. Blue cells favor the refined statistic and pink cells favor the counterfactual statistic; both panels share one symmetric color scale. **e,** Nominal false-positive rates at $t = 10{,}000$. **f,** Formula-specific empirical 95th-percentile thresholds trained on an independent half-null sample.

Decision changes occur in short alignments

The simulation curves include the JC control and the two Tree-of-Life rate-heterogeneous models in the true-tree fixed-length scenario. Under JC, the published and weighted curves coincide, consistent with the repeated non-stationary eigenvalue. Under GTR+F+G4 and LG+G4, decision changes are concentrated in short alignments and long target branches. At 100 sites, the weighted curves remain above the published SatuTe curves at branch lengths 10–12. At 250 sites the separation is smaller, and at 1,000 sites the curves nearly coincide.

The 1,000-replicate grid gives the same interpretation. In the true-tree fixed-length scenario at 100 sites under GTR+F+G4, eigenvalue weighting increased the informative-call fraction by 16.9 percentage points at branch length 10 in the five-taxon case. At branch length 12, the increase was 20.9 percentage points. In the 16-taxon case, the corresponding changes were 22.6 percentage points at both branch lengths. Under LG+G4, the changes were 24.9 and 32.2 percentage points in the five-taxon case. In the 16-taxon case, they were 26.9 and 35.5 percentage points. At 250 sites, the largest changes across these model and tree combinations were 6.5–11.1 percentage points. At 1,000 sites, the two formulas almost always gave the same binary decision.

Decision changes are localized

The branch-neighborhood analysis tests whether the decision changes are specific to the focal branch. In the 1,000-replicate grid, the difference was concentrated on the target branch. In the true-tree fixed-length scenario at 100 sites and branch length 12, the target-branch change was 14.5 percentage points in the five-taxon GTR+F+G4 case. In the 16-taxon GTR+F+G4 case, it was 22.0 percentage points. The nearest-neighbor change was zero in both cases. Under LG+G4, the corresponding target-branch changes were 38.1 and 39.1 percentage points, again with zero nearest-neighbor change. In these simulations, the weighted statistic changed calls on the tested branch rather than across the surrounding tree.

Biological sliding-window rerun

The biological examples use the Tree-of-Life panels from the original SatuTe paper. The protein analysis uses the ribosomal-protein 2D tree, and the nucleotide analysis uses the 16S rRNA 3D tree. In both trees, the highlighted internal split is the branch leading to Eukaryota. The yeast branch is an external branch within the Eukaryota subtree and is the second focal branch in the sliding-window analyses.

The curated biological rerun applies the same comparison to the 16S rRNA Tree-of-Life dataset shown in the 3D tree panel. The dataset has 1,947 alignment sites, a fitted GTR+F+G4 model, four rate categories and 36-site sliding windows. The comparison includes the published SatuTe rerun, a native SatuTe recomputation using $C_{\partial 1}$, and a native eigenvalue-weighted recomputation. Differences between the published and native SatuTe curves reflect differences in implementation and input files. The formula comparison is therefore the native SatuTe recomputation versus the native eigenvalue-weighted recomputation.

For the branch leading to Eukaryota, the published SatuTe rerun marks 822 of 1,912 windows as saturated. The native SatuTe recomputation marks 895 windows, whereas the native eigenvalue-weighted recomputation marks 699 windows. For the yeast branch, the corresponding counts are 70, 83 and 29 saturated windows. The published-to-native difference is not used as a formula-effect estimate.

A second biological rerun uses an independent implementation that does not read SatuTe component files or SatuTe result trees. It recomputes the sliding-window scores from the IQ-TREE treefile, model report, site-probability output and original alignments for both Tree-of-Life examples [4]. The z-score curves show the same branch order and window size as the published

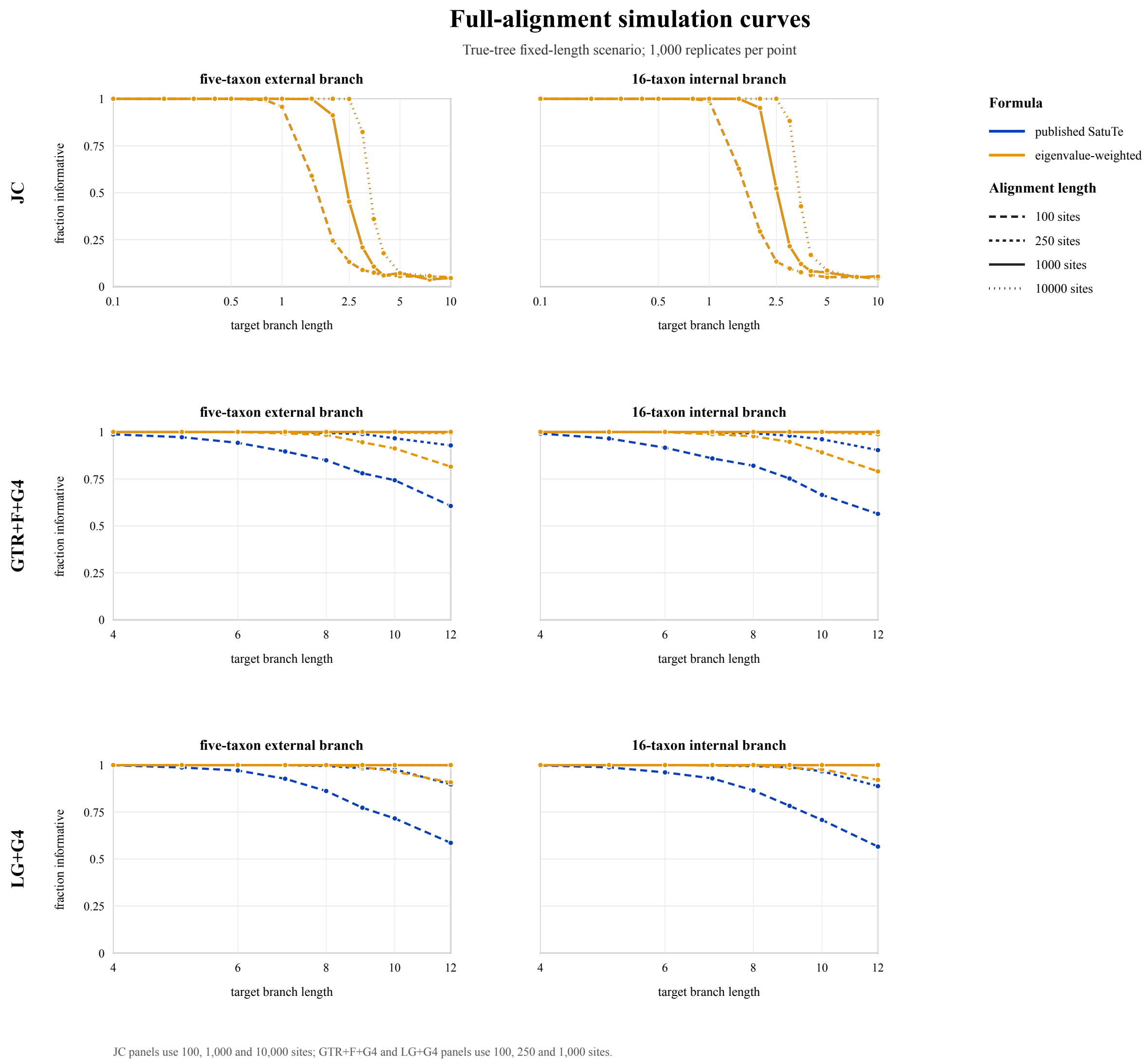

Figure 3: **Full-alignment simulation curves with and without rate heterogeneity.** Curves show the fraction of replicates called phylogenetically informative in the true-tree fixed-length scenario. The JC panels use 100, 1,000 and 10,000 sites; the GTR+F+G4 and LG+G4 panels use 100, 250 and 1,000 sites. Under JC, the published SatuTe and eigenvalue-weighted curves overlap.

sliding-window analysis. Because this rerun contrasts published outputs with an independent recomputation, the curves are not used as the primary formula-effect estimate.

In the protein 2D tree, the branch leading to Eukaryota has 2,561 windows. The saturated-window counts are 138 in the published SatuTe rerun and 5 in the native eigenvalue-weighted recomputation. The yeast branch also has 2,561 windows, with counts 54 and 8. In the 16S rRNA 3D tree, the branch leading to Eukaryota has 1,912 windows, with counts 822 and 699. The yeast branch has 1,912 windows, with counts 70 and 29. These counts summarize the threshold crossings in the sliding-window curves.

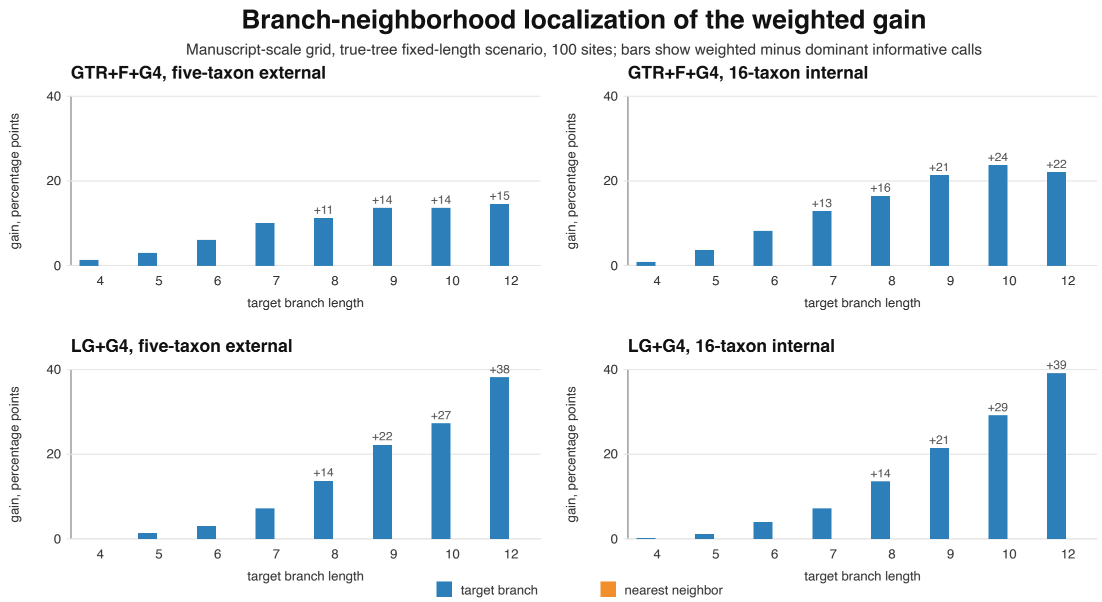

Figure 4: **Branch-neighborhood localization in the 1,000-replicate rate-heterogeneous grid.** Bars show the change in informative-call fraction from eigenvalue weighting at 100 sites in the true-tree fixed-length scenario. The main decision changes occur on the target branch; nearest-neighbor changes are zero in these cells.

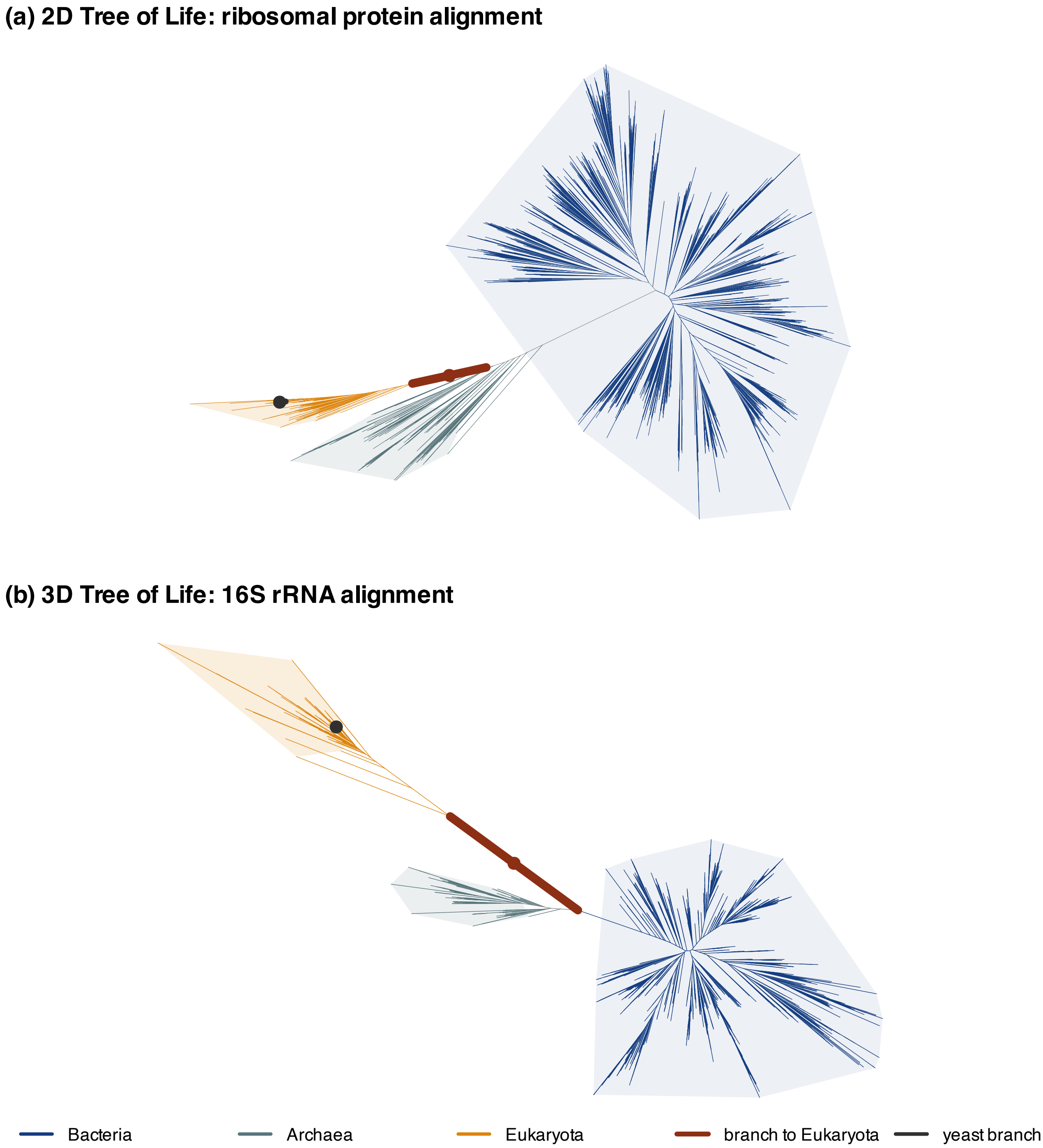

Figure 5: **Tree-of-Life references for the biological reruns. a,** Ribosomal-protein 2D tree. **b,** 16S rRNA 3D tree. Branch colors and translucent hulls identify Bacteria, Archaea and Eukaryota. The highlighted internal segment marks the branch leading to Eukaryota, and the black marker identifies the yeast branch used as the external-branch reference.

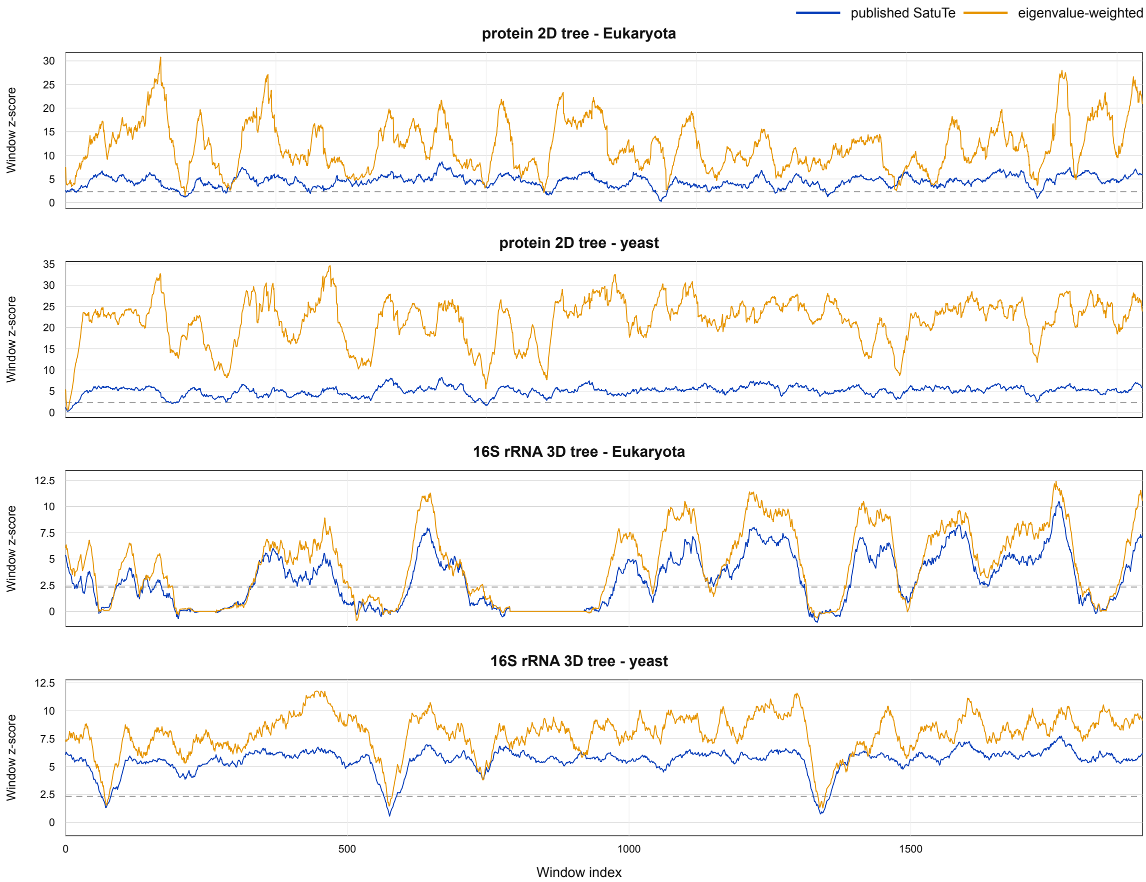

Figure 6: **Published SatuTe and independent eigenvalue-weighted sliding-window curves in the Tree-of-Life examples.** Curves show 36-site window Z-scores for the branch leading to Eukaryota and for the yeast branch in the ribosomal-protein 2D tree and the 16S rRNA 3D tree. The dashed line marks z_α.

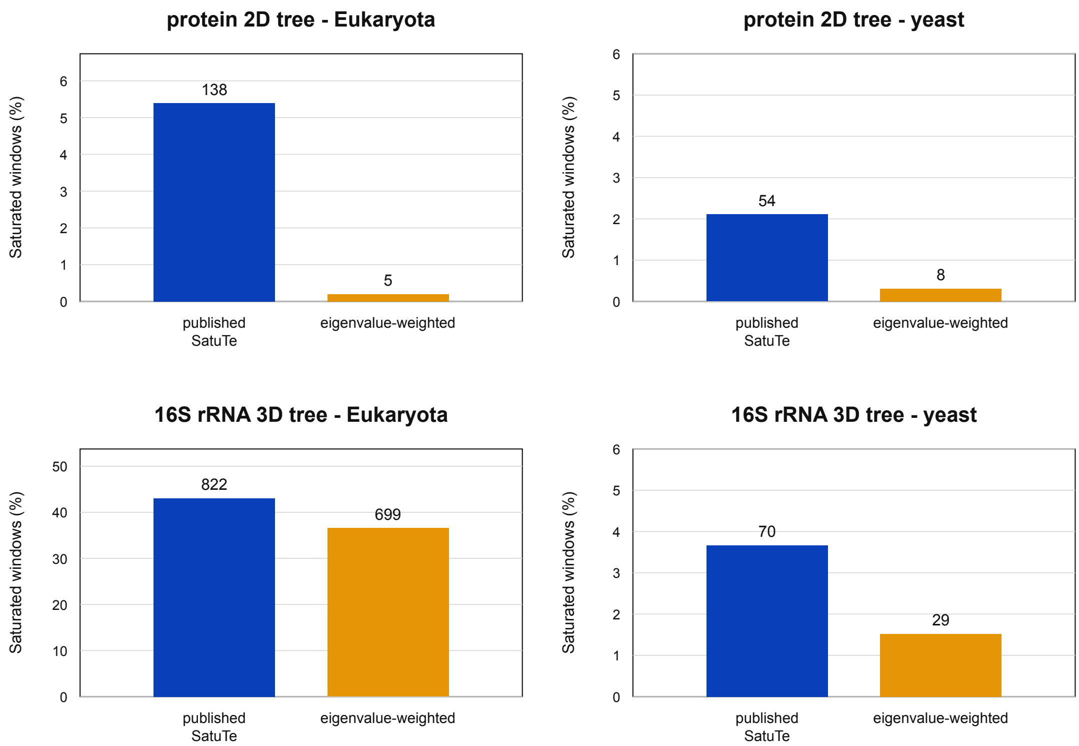

Figure 7: **Published SatuTe and independent eigenvalue-weighted sliding-window calls in the Tree-of-Life examples.** Bars show the fraction of windows classified as saturated for the branch leading to Eukaryota and for the yeast branch. Numbers above bars give saturated-window counts.

Discussion

The weighted statistic changes one step of SatuTe. The branch, subtree likelihood vectors, null hypothesis and one-sided test remain the same. When the fitted substitution model separates the other non-stationary components, the statistic includes them after scaling by their expected decay along branch AB.

The simulations identify the settings in which the formula changes the decision. Under JC, the statistic is unchanged. Under four-category rate-heterogeneous GTR and LG models, the weighted statistic changed decisions most often for 100-site alignments and long tested branches. With more sites, the published SatuTe statistic usually had enough information to call the branch phylogenetically informative, and the weighted statistic rarely changed the binary decision. These results do not support a general claim of higher power; they mark the tested settings in which the formula difference changes the decision.

The explicit invariant-site analysis identifies a separate issue. A zero-rate category never loses the shared ancestral state across the focal branch, so treating that category as independent in the infinite-branch null creates a positive score even when the data are generated from the correct null. Studentization then amplifies the shift at rate $\sqrt{n}$, explaining why the counterfactual nominal rejection rate approaches one as alignment length grows. The refined denominator restores exact null centering and ordinary standard-normal calibration. Once both formulas are given empirical thresholds, the remaining power change is smaller and localized to the detection boundary, where the refined statistic gained as much as 16.3 percentage points in the tested grid.

The dense grid also clarifies resolution. A coarse set of branch lengths captures the false-positive failure but understates the peak power difference because the detection transition moves with alignment length and rate mixture. The 2.5-unit steps from 10 to 25 and 5-unit steps from 30 to 50 resolve the diagonal power-gain ridge in the full-grid figure. Outside this ridge, both formulas usually have either near-null or near-complete power, so a formula change has little room to alter the binary decision.

The branch-neighborhood analysis addresses branch specificity. If the weighted statistic simply made all branches easier to call informative, the same change would appear on neighboring branches. In the 1,000-replicate simulations, the decision changes were concentrated on the tested branch, while nearest neighbors were already informative under both formulas. This is consistent with a branch-specific interpretation within the simulated five-taxon and 16-taxon settings.

In the Tree-of-Life $+G4$ examples, the recomputations changed sliding-window saturation calls. These examples do not by themselves quantify the formula effect, because the published and native curves also differ in input files and implementation. The paired full-alignment simulations therefore provide the more controlled test. The sliding-window results are biological examples of the same question: whether regions of an alignment retain enough phylogenetic signal for a selected branch.

The original SatuTe paper did not resolve the true phylogenetic relationship of Archaea, Bacteria and Eukaryota, and this extension does not change that scope. SatuTe tests for saturation; it does not determine which tree is correct. The weighted statistic keeps that role. It applies when a rate-heterogeneous model has separated non-stationary eigenvalues and the analysis asks whether signal outside the second-largest-eigenvalue component should contribute. The published SatuTe coefficient remains the simpler reference statistic.

Online Methods

Notation

We use the notation of SatuTe. A branch AB separates subtrees T_A and T_B. A site pattern ∂ induces subpatterns ∂_A and ∂_B. The partial likelihood vectors at the endpoints are $L(\partial_A)$

and $L(\partial_B)$. The stationary distribution is π, with $P(\partial_A) = \langle L(\partial_A), \pi\rangle$ and $P(\partial_B) = \langle L(\partial_B), \pi\rangle$. The reversible rate matrix has eigenvalues $\lambda_0 = 0 > \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_d$ and left eigenvectors h_i, where d is the number of non-stationary components. Coherence coefficients are defined by Eq. (1).

Weighted estimator

For each site s, $C_{\partial_s i}$ is computed for the non-stationary modes. For branch length t, the mode weight is $w_i(t) = \exp\{(\lambda_i - \lambda_1)t\}$. The quantities $\hat{C}_\lambda(t)$, $\hat{\sigma}^2_\lambda(t)$ and $Z_\lambda(t)$ are then computed using Eqs. (4) to (6). If a model has a repeated second largest eigenvalue, each direction in that eigenspace has weight one. For JC, the estimator is the published SatuTe statistic.

Null distribution

The weighted statistic keeps the SatuTe null hypothesis. Let

$$X_i(\partial_A) = \left\langle \frac{L(\partial_A)}{P(\partial_A)}, h_i \right\rangle, \qquad Y_i(\partial_B) = \left\langle \frac{L(\partial_B)}{P(\partial_B)}, h_i \right\rangle.$$

Then $C_{\partial i} = X_i Y_i$ and

$$C^\lambda_\partial(t) = \sum_{i=1}^{d} w_i(t) X_i Y_i.$$

Under the SatuTe null hypothesis, the two subtree patterns ∂_A and ∂_B are independent. The non-stationary eigenvectors are orthogonal to the stationary component, so the original SatuTe argument gives $\mathbb{E}[X_i] = 0$ and $\mathbb{E}[Y_i] = 0$. Therefore $\mathbb{E}[C_{\partial i}] = 0$ for every non-stationary component, and by linearity $\mathbb{E}[C^\lambda_\partial(t)] = 0$.

The variance follows by expanding the square of the weighted sum. Define the subtree second moments

$$\sigma^2_{A,i,j} = \mathbb{E}[X_i X_j], \qquad \sigma^2_{B,i,j} = \mathbb{E}[Y_i Y_j].$$

Using independence of the two subtrees under the null,

$$\mathrm{Var}\Big(C^\lambda_\partial(t)\Big) = \sum_{i=1}^{d}\sum_{j=1}^{d} w_i(t) w_j(t) \sigma^2_{A,i,j} \sigma^2_{B,i,j}. \tag{12}$$

For n independent sites, $\mathrm{Var}[\hat{C}_\lambda(t)] = \mathrm{Var}(C^\lambda_\partial(t))/n$, which is the population form estimated in Eq. (5). With finite fourth moments and a consistent variance estimator, the central limit theorem and Slutsky's theorem give $Z_\lambda(t) \xrightarrow{d} N(0,1)$ under subtree independence. Maximum-likelihood topology selection remains a separate model-selection step, so the same Bonferroni correction used by SatuTe is retained for that scenario.

Simulation protocol

The rate-heterogeneous simulations use full alignments and paired formula evaluation. The five-taxon tree is a balanced four-taxon subtree T_A with all subtree branches set to 0.2 and a single taxon B attached by branch AB. The 16-taxon tree is built from two balanced eight-taxon subtrees. Each alignment is analyzed under four scenarios. Two scenarios use the true topology: one with fixed true branch lengths and one with maximum-likelihood branch lengths. The other two use the maximum-likelihood inferred tree, without or with the Bonferroni threshold $\alpha/(m_A m_B)$. Within each replicate, both formulas are evaluated on the same tree, branch length and target split.

The nucleotide simulations use the Tree-of-Life 16S rRNA GTR+F+G4 setting, implemented as the corresponding fixed four-category rate model used by the IQ-TREE integration [4].

The protein simulations use the Tree-of-Life LG+G4 setting. The growing-site grid uses $100, 250, 500, 1{,}000, 2{,}000, 4{,}000$ and $10{,}000$ sites. It uses target branch lengths $4, 5, 6, 7.5, 10$ and 12, with 100 replicates per condition. The 1,000-replicate grid uses $100, 250$ and $1{,}000$ sites, target branch lengths $4, 5, 6, 7, 8, 9, 10$ and 12. The branch-neighborhood analysis computes the same dominant and weighted statistics for the target branch and adjacent branches in the same simulated trees.

Paired invariant-mixture grid

The $+I$ analysis uses a four-taxon tree with two two-tip subtrees, terminal branches of length 0.05 and a varied focal branch. The substitution process is fixed to GTR with exchangeabilities $\{1, 2, 1, 1, 2, 1\}$ and stationary frequencies $\{0.30, 0.20, 0.20, 0.30\}$. Both models have invariant proportion $q_0 = 0.10$. The $+I + G4$ model uses four equal-probability discrete-gamma categories with shape 0.5. The $+I+R4$ model uses positive-category proportions $\{0.35, 0.25, 0.23, 0.17\}$ and rates $\{0.0747656, 0.420849, 1.15850, 3.54214\}$, rescaled after addition of the invariant component so that the overall mean rate is one.

All $4^4 = 256$ site-pattern probabilities are enumerated exactly under the correct finite-branch distribution and under the infinite-branch null in Eq. (8). Each simulated alignment is a multinomial draw from one of these exact distributions. The same pattern counts are evaluated by the counterfactual and refined statistics, producing paired decisions without simulator-stream variation.

Alignment lengths are

$$n \in \{100, 500, 1000, 5000, 10000\}. \tag{13}$$

The dense focal-branch grid is

$$\begin{aligned} t \in \{&0.5, 1, 2, 3, 4, 5, 7.5, 10, 12.5, 15, 17.5, 20, 22.5, 25, \\ &30, 35, 40, 45, 50, 60, 75, 100, 150, 200, 500, 1000, 10000\}. \end{aligned} \tag{14}$$

Thus the grid uses 2.5-unit resolution from 10 to 25 and 5-unit resolution from 30 to 50, where the detection transitions occur, while retaining short and extreme controls. For every model, branch length and alignment length, 20,000 null and 20,000 finite-branch alternative alignments are drawn. Half of the null replicates train a formula-specific empirical 95th-percentile threshold; the other half evaluates held-out false-positive control. Power is measured on the independent alternative sample. Wilson intervals quantify individual rejection fractions. Paired percentile-bootstrap intervals use 4,000 resamples of the three discordance outcomes: refined only, agreement and counterfactual only.

Biological sliding-window reruns

The biological reruns use the same 36-site window size as the original SatuTe Tree-of-Life analysis. The curated rRNA rerun compares the published SatuTe result, a native SatuTe recomputation and a native eigenvalue-weighted recomputation for the Eukaryota and yeast branches in the 16S rRNA 3D tree. The independent rerun recomputes scores from the IQ-TREE treefile, model report, site-probability output and original alignments for both the ribosomal-protein 2D tree and the 16S rRNA 3D tree. Saturated windows are windows with $Z \le z_\alpha$, following the SatuTe decision rule used for sliding-window analyses.

Data availability

The repository contains reviewed summaries under `artifacts/releases/`, including the dense invariant-mixture release and curated biological sliding-window evidence. Raw LiSC shards and workstation runs are stored under ignored `artifacts/cluster/` and `artifacts/local/` directories. The invariant-mixture analysis is reproduced by `experiments/003_invariant_mixture/run.py`.

Code availability

The IQ-TREE integration, plotting scripts and manuscript sources are in the repository. The repository also contains the scripts used to run the paired full-alignment and exact-pattern invariant-mixture simulations and to produce the merged summaries, high-resolution figures and branch-neighborhood outputs.

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Author contributions

B.S. developed the weighted-statistic formulation and prepared the manuscript.

Competing interests

The author declares no competing interests.

References

[1] Manuel, C., Sakalli, E., Schmidt, H. A., Viñas, C., von Haeseler, A. & Elgert, C. When the past fades: detecting phylogenetic signal with SatuTe. *Mol. Biol. Evol.* **42**, msaf090 (2025).

[2] Levin, D. A. & Peres, Y. *Markov Chains and Mixing Times.* 2nd edn. American Mathematical Society (2017).

[3] Tavaré, S. Some probabilistic and statistical problems in the analysis of DNA sequences. *Lect. Math. Life Sci.* **17**, 57–86 (1986).

[4] Minh, B. Q. et al. IQ-TREE 2: new models and efficient methods for phylogenetic inference in the genomic era. *Mol. Biol. Evol.* **37**, 1530–1534 (2020).